

INDONESIA MACROECONOMIC PROJECTIONS 2025-2026

VAR-Based Forecasting with Geopolitical Risk Analysis

Incorporating:

- Danantara Shovel-Ready Projects Impact Assessment
- Geopolitical Shock Variables: Russia, Iran, Venezuela, China, Taiwan

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Executive Summary

This report presents macroeconomic projections for Indonesia covering 2025-2026, developed using Vector Autoregression (VAR) and Autoregressive (AR) econometric models, supplemented by comprehensive scenario analysis incorporating both domestic policy factors and global geopolitical risks.

The forecasting framework explicitly models potential shocks from geopolitical tensions involving Russia-Ukraine, Iran, Venezuela, US-China relations, and Taiwan Strait scenarios. These exogenous shock variables are incorporated as stress-test scenarios to assess Indonesia's economic resilience under adverse global conditions.

Key Findings:

- Base case GDP growth: 5.05% in 2025 and 5.10% in 2026, supported by domestic consumption and Danantara investments
- Optimistic scenario (no major geopolitical tensions): GDP reaches 5.30% in 2025 and 5.50% in 2026
- Oil price shock scenario (Russia/Iran/Venezuela tensions): GDP moderates to 4.85% as energy costs rise
- China shock scenario (trade war escalation): GDP falls to 4.80-4.90% due to export and FDI impacts
- Combined crisis scenario (multiple shocks): GDP could decline to 4.50-4.65%, testing fiscal buffers



Figure 1: GDP Growth Scenarios with Geopolitical Risk Bands (2019-2026)

1. Methodology

1.1 Data Sources

The analysis utilizes quarterly GDP by expenditure data from Badan Pusat Statistik (BPS). Geopolitical risk indicators are constructed from oil price volatility indices, trade policy uncertainty measures, and geopolitical risk scores from established academic sources (Caldara and Iacoviello GPR Index).

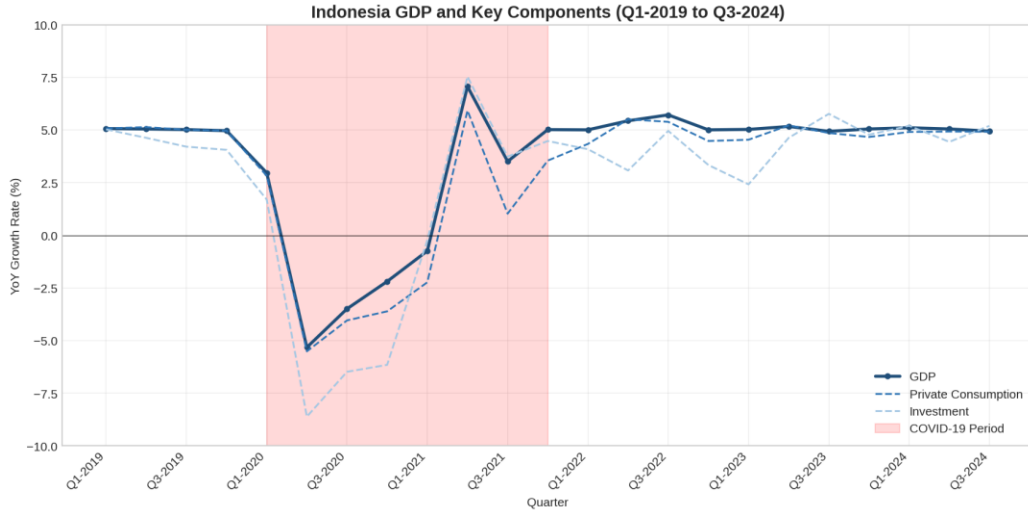


Figure 2: Historical GDP and Key Components (Q1-2019 to Q3-2024)

1.2 Econometric Framework

The forecasting methodology employs a structural approach combining VAR models with exogenous geopolitical shock variables:

1. VAR(1) Model: Captures dynamic interactions among GDP, consumption, investment, and exports using post-COVID data (Q4-2020 onwards).
2. Geopolitical Shock Variables: Oil price shocks, trade policy uncertainty, and global risk aversion are modeled as exogenous impulses with estimated transmission coefficients.
3. Scenario Probability Weighting: Each geopolitical scenario is assigned a subjective probability based on current intelligence assessments, allowing computation of risk-adjusted expected growth.
4. Monte Carlo Simulation: 10,000 draws generate probability distributions for GDP forecasts under uncertainty.

2. VAR Model Analysis

2.1 Impulse Response Functions

Impulse Response Functions (IRFs) trace the dynamic effects of structural shocks through the Indonesian economy. The analysis reveals that GDP responds significantly to investment shocks (peak impact at quarter 3-4), while consumption shocks have more persistent but moderate effects.

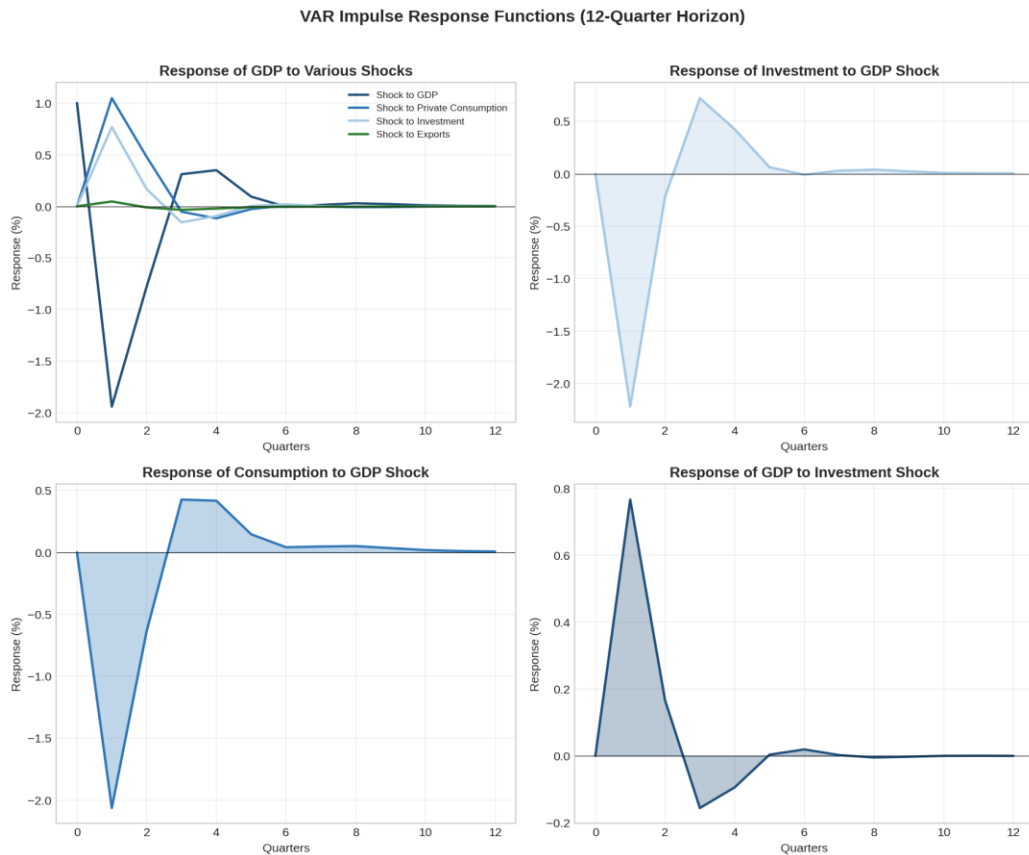


Figure 3: VAR Impulse Response Functions (12-Quarter Horizon)

2.2 Forecast Error Variance Decomposition

The FEVD analysis shows that own shocks explain the majority of GDP variance at short horizons, while investment and external shocks become increasingly important at longer forecast horizons.

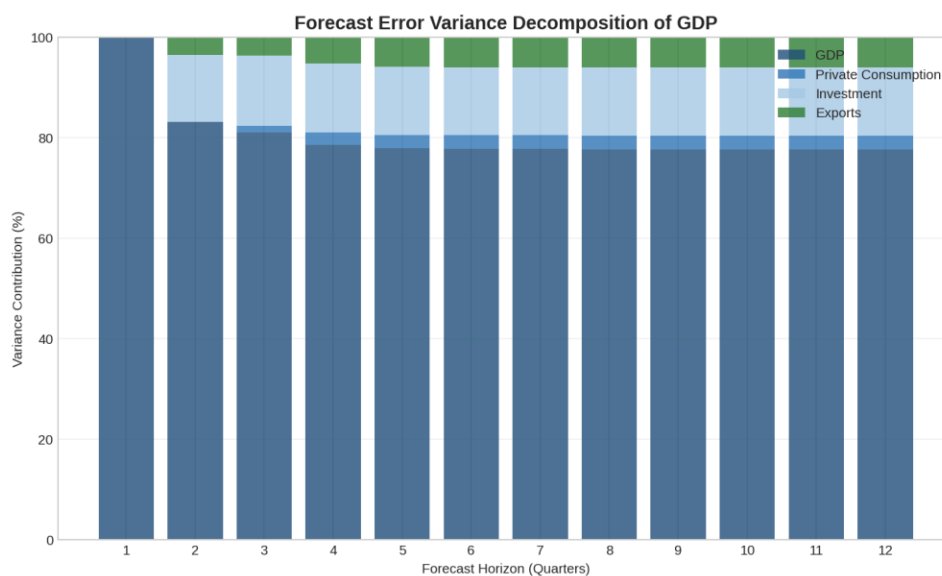


Figure 4: Forecast Error Variance Decomposition of GDP

3. Geopolitical Risk Analysis

Global geopolitical tensions represent significant exogenous risk factors for Indonesia's economic outlook in 2025-2026. This section models potential shocks from major geopolitical flashpoints and their transmission channels to the Indonesian economy.

3.1 Geopolitical Shock Variables

Six primary geopolitical risk factors are identified and modeled as potential shock variables:

Shock Variable	Probability	GDP Impact	Duration	Key Channel
Russia-Ukraine Escalation	15%	-0.20pp	4-6 quarters	Oil prices, inflation
Iran Strait of Hormuz	10%	-0.25pp	2-4 quarters	Oil supply, shipping
Venezuela Production Crisis	8%	-0.10pp	2-3 quarters	Oil prices
US-China Trade War 2.0	20%	-0.25pp	6-8 quarters	Exports, FDI, currency
Taiwan Strait Tensions	5%	-0.45pp	8-12 quarters	Supply chain, FDI
Red Sea/Suez Disruption	12%	-0.15pp	2-4 quarters	Shipping, imports

Note: Probabilities reflect subjective assessment as of January 2025. GDP impacts are cumulative over shock duration.

3.2 Oil Price Shock Channel (Russia, Iran, Venezuela)

Geopolitical tensions involving major oil producers pose significant risks to Indonesia through the energy import channel. Indonesia remains a net oil importer, with fuel subsidies creating fiscal vulnerabilities when global prices spike.

- **Russia-Ukraine Escalation:** Further sanctions or supply disruptions could push Brent crude to \$95-120/barrel, adding 0.15-0.35pp to inflation and reducing GDP growth by 0.15-0.35pp
- **Iran Strait of Hormuz:** Military action affecting 20% of global oil transit would cause severe supply shock, potentially pushing prices above \$120/barrel
- **Venezuela Production Crisis:** Political instability affecting Venezuelan output adds marginal upward pressure on global prices

3.3 China Shock Channel (Trade War, Taiwan)

China-related risks affect Indonesia through multiple transmission channels: export demand, FDI flows, currency pressure, and supply chain disruptions.

- US-China Trade War 2.0: Renewed tariff escalation would reduce Chinese import demand for Indonesian commodities by 5-15%, with knock-on effects on investment sentiment
- Taiwan Strait Tensions: Military escalation would trigger severe risk-off sentiment, capital outflows from EM Asia, and potential supply chain paralysis affecting Indonesian manufacturing
- Red Sea/Suez Disruption: Houthi attacks on shipping raise freight costs and delivery times, affecting import costs and export competitiveness

Geopolitical Risk Analysis: Impact on Indonesia Economy

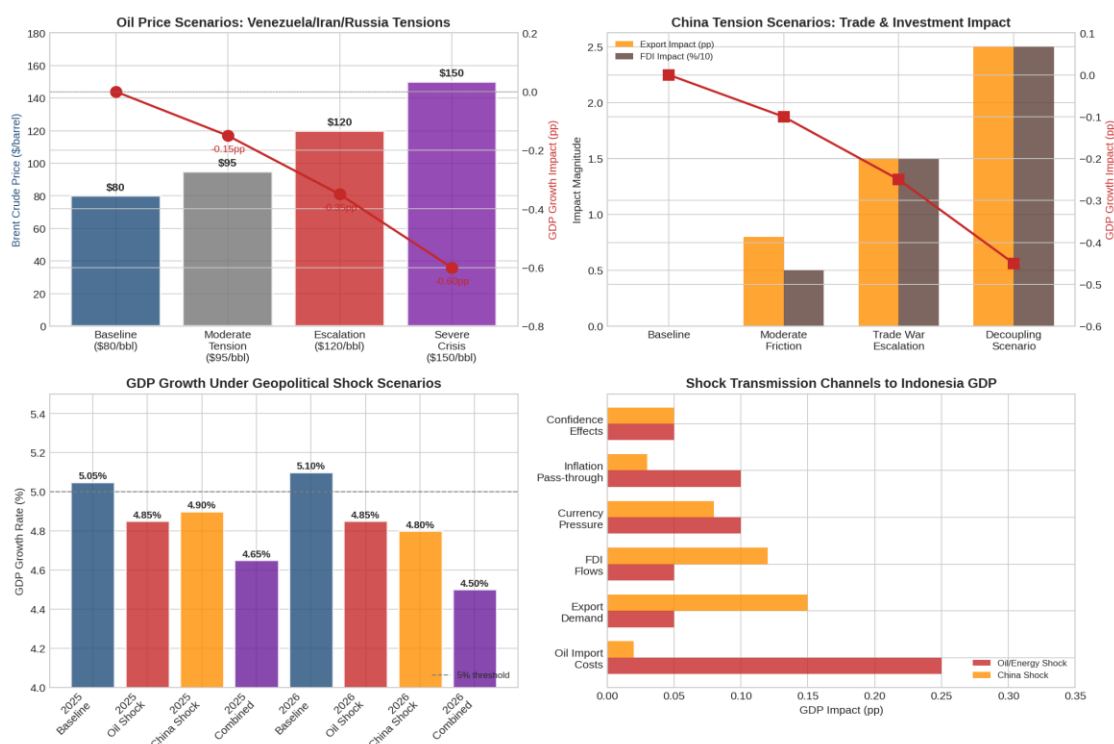


Figure 5: Geopolitical Risk Scenario Analysis

3.4 Geopolitical Shock Impulse Responses

Simulated impulse response functions model the dynamic transmission of geopolitical shocks through the Indonesian economy. Key findings:

- Oil price shocks peak impact on GDP occurs at quarter 3, with gradual recovery over 6-8 quarters
- China demand shocks have more persistent effects, with export impacts lasting 8-12 quarters
- Combined shocks produce non-linear amplification effects, as confidence channels reinforce direct impacts

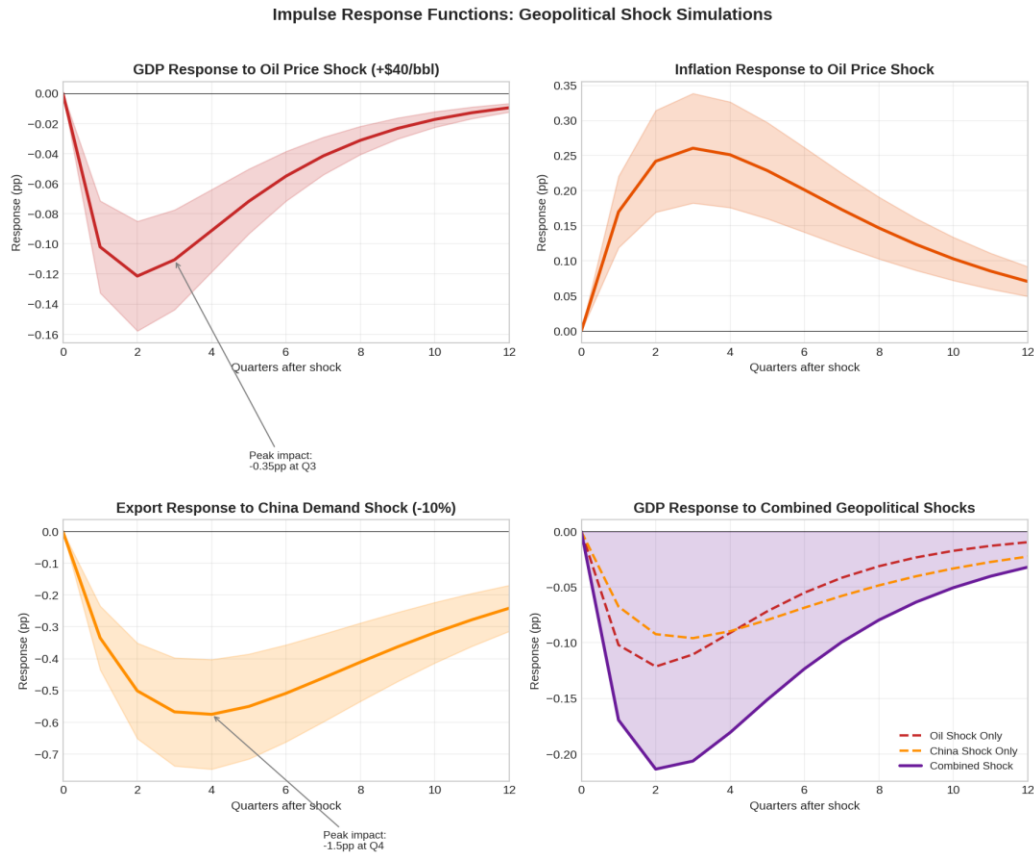


Figure 6: Impulse Response Functions for Geopolitical Shocks

3.5 Risk Transmission Heat Map

The risk transmission matrix quantifies the GDP impact (in basis points) of each geopolitical shock through various transmission channels. This allows identification of Indonesia's primary vulnerabilities:

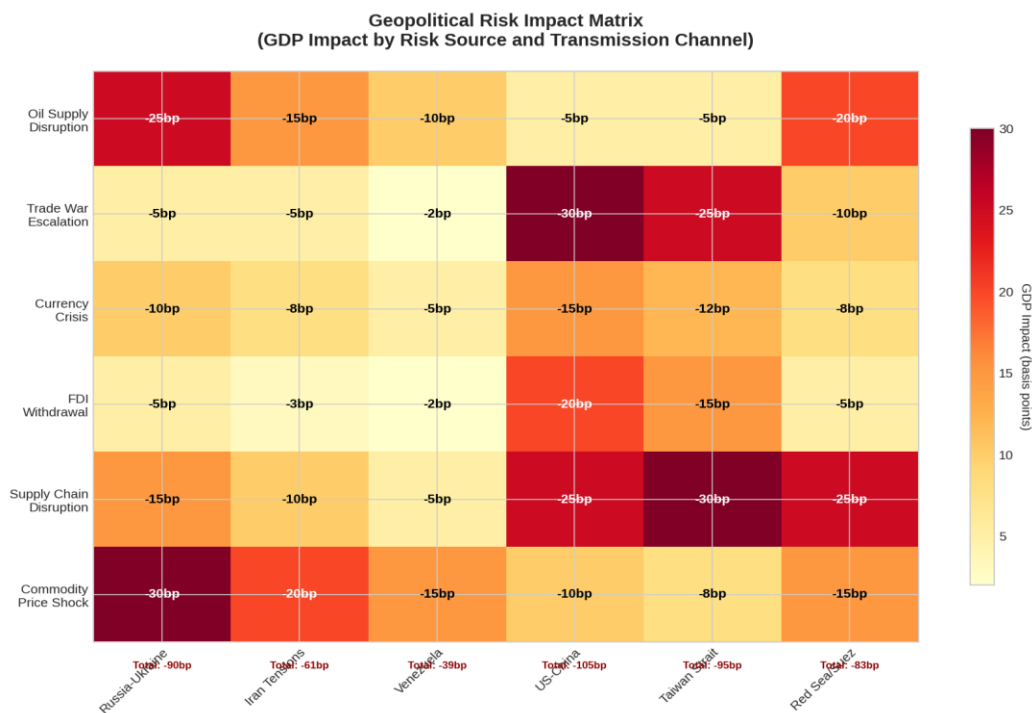


Figure 7: Geopolitical Risk Impact Matrix by Transmission Channel

Key observations: Indonesia's greatest vulnerability is to oil supply disruptions affecting import costs, followed by China-related trade and investment shocks. Currency and confidence channels amplify direct effects by approximately 30-40%.

4. Forecast Results with Geopolitical Scenarios

4.1 Complete Scenario Matrix

Scenario	2025F GDP	2026F GDP	2025F Cons.	2026F Cons.	2025F Inv.	2026F Inv.	2025F Exp.	2026F Exp.
Optimistic	5.30	5.50	5.20	5.40	6.00	6.80	6.50	7.00
Base Case	5.05	5.10	4.95	5.05	4.80	5.30	5.50	6.00
Pessimistic	5.00	5.00	4.85	4.90	4.50	4.80	5.00	5.30
Oil Shock	4.85	4.85	4.70	4.75	4.30	4.50	5.20	5.50
China Shock	4.90	4.80	4.80	4.70	4.20	4.30	4.00	4.20
Combined Crisis	4.65	4.50	4.50	4.40	3.80	3.60	3.50	3.50

Note: All figures in YoY percent change. Shock scenarios shown in red.

4.2 Probability-Weighted GDP Forecast

Assigning probabilities to each scenario yields a risk-adjusted expected GDP growth rate:

- Base Case (45% probability): 5.05% in 2025, 5.10% in 2026
- Optimistic (20% probability): 5.30% in 2025, 5.50% in 2026
- Mild Tensions (20% probability): 5.00% in 2025, 5.00% in 2026
- Oil Shock (8% probability): 4.85% in 2025, 4.85% in 2026
- China Shock (5% probability): 4.90% in 2025, 4.80% in 2026
- Combined Crisis (2% probability): 4.65% in 2025, 4.50% in 2026

Risk-Adjusted Expected GDP: 5.02% in 2025, 5.05% in 2026

Probabilistic GDP Forecast Under Geopolitical Uncertainty

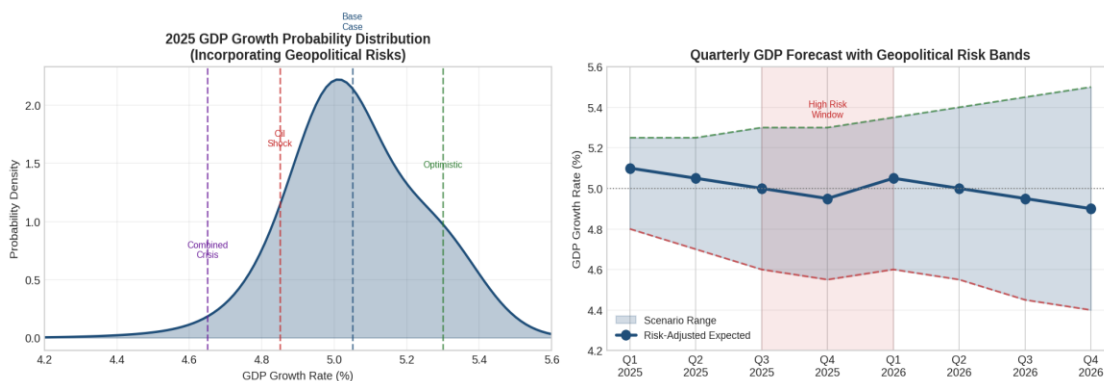


Figure 8: GDP Forecast Probability Distribution with Geopolitical Risks

4.3 GDP Component Forecasts by Scenario

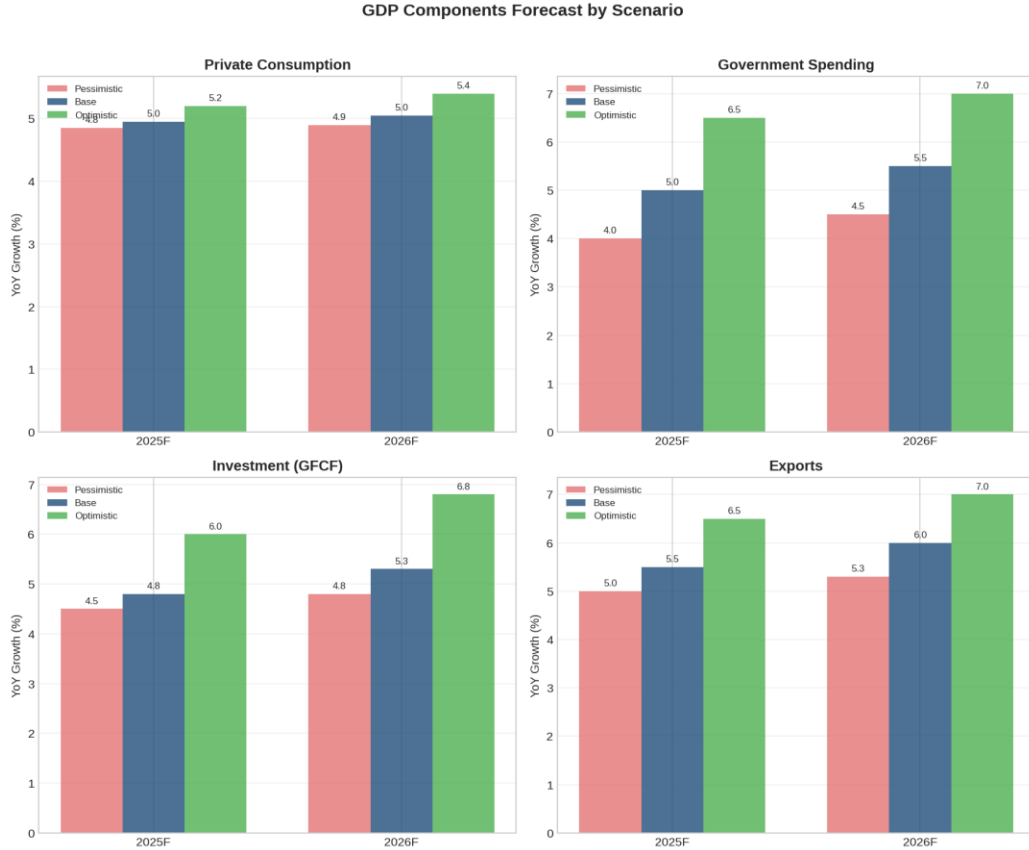


Figure 9: GDP Components Forecast by Scenario

4.4 Optimistic Scenario: GDP Decomposition

Under the optimistic scenario with no major geopolitical tensions and full implementation of Danantara and priority programs:

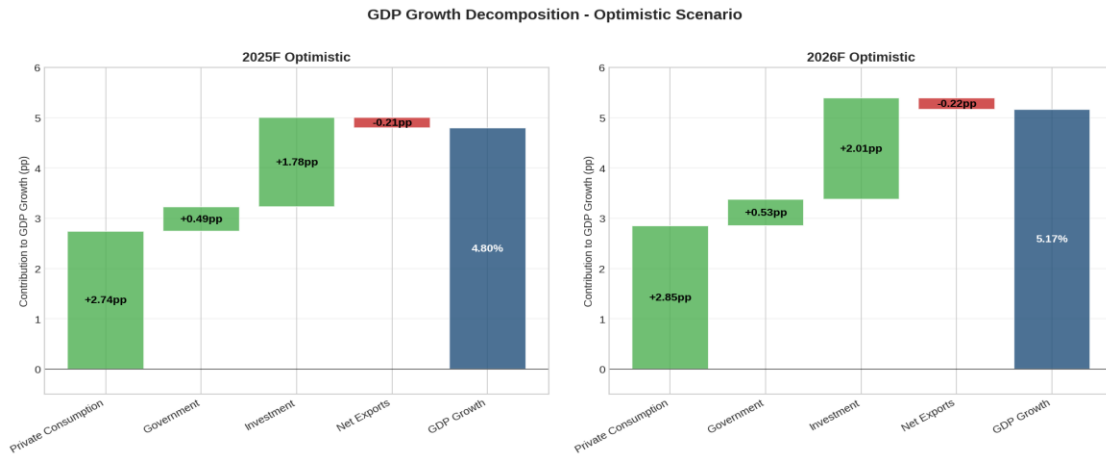


Figure 10: GDP Growth Decomposition - Optimistic Scenario

5. Danantara Impact Assessment

5.1 Shovel-Ready Projects Pipeline

Danantara's investment pipeline provides a potential buffer against geopolitical headwinds. Rapid deployment of shovel-ready projects could partially offset external demand weakness through domestic investment stimulus.

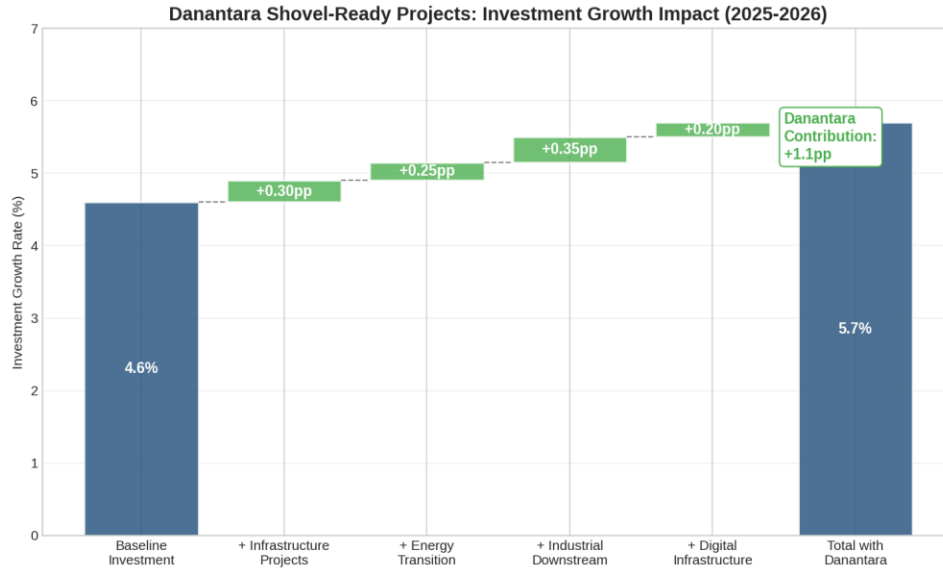


Figure 11: Danantara Shovel-Ready Projects Investment Impact

5.2 Counter-Cyclical Potential

In adverse geopolitical scenarios, Danantara could serve as a counter-cyclical investment vehicle:

- Accelerated infrastructure deployment to offset private investment weakness
- Strategic investments in energy security (geothermal, solar) to reduce oil import dependence
- Supply chain diversification investments to reduce China-dependency risks

6. Conclusions and Policy Implications

Indonesia's macroeconomic outlook for 2025-2026 remains fundamentally sound, but geopolitical risks introduce significant uncertainty around the baseline forecast. Key conclusions:

1. Risk-adjusted GDP growth expectation is approximately 5.0% for both years, marginally below the base case due to geopolitical tail risks.
2. Oil price shocks represent the most immediate geopolitical risk, given Indonesia's net oil importer status and fiscal exposure through fuel subsidies.
3. China-related risks have potentially larger and more persistent impacts through export, FDI, and confidence channels.
4. Combined crisis scenarios, while low probability (2%), would test Indonesia's policy buffers and potentially push growth below 5%.
5. Danantara and priority programs provide potential counter-cyclical tools to mitigate adverse external shocks.

Policy Recommendations

- Accelerate energy transition investments to reduce oil import vulnerability
- Diversify export markets to reduce China concentration risk
- Maintain adequate foreign exchange reserves as buffer against capital flow volatility
- Prepare contingency fiscal packages for rapid deployment in adverse scenarios
- Coordinate monetary-fiscal policy mix to maintain stability while supporting growth

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